Zhaoxiang (Simon) Cai

PhD (Cancer Data Science) · MBusA (First Class Honours) · BCompSc (First Class Honours)

Senior Data Scientist (Cancer Data Science), Children's Medical Research Institute
Adjunct Lecturer, Faculty of Medicine and Health, University of Sydney

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CAREER PROFILE

I am a cancer data scientist specialising in artificial intelligence and machine learning for large-scale molecular data analysis. I hold a first-class honours degree in Computer Science from Monash University, where I graduated as the top student, and a PhD in Cancer Data Science from the University of Sydney. After gaining software engineering experience at Goldman Sachs, I transitioned to cancer research and now serve as an Adjunct Lecturer at the University of Sydney and Senior Data Scientist at the Children's Medical Research Institute (CMRI).

My research focuses on developing and applying deep learning methods to cancer multi-omics data, with particular expertise in federated learning, generative models, and transformer architectures. As co-first author on the landmark Cancer Cell study mapping the proteomic landscape of 949 human cancer cell lines, I contributed to one of the field's most widely used resources (260+ citations). My subsequent first-author publications in Cancer Discovery and Nature Communications have introduced novel approaches in federated deep learning for privacy-preserving cancer subtyping, and generative AI for synthetic augmentation of multi-omic datasets. I maintain active international collaborations with the Wellcome Sanger Institute (UK) and the University of Lisbon (Portugal).

In 2025, I was awarded a Cancer Institute NSW Early Career Fellowship ($597,732 over three years) to develop proteomic foundation models integrated with federated learning for multi-hospital cancer research. My work to date has generated over 1,160 citations (h-index: 8) and I have contributed to more than $2.59 million in competitive research funding. I am driven by the goal of translating advanced computational methods into practical tools that improve cancer diagnosis and prognosis through precision medicine.

1,166	8	8	5 of 10
TOTAL CITATIONS	H-INDEX	I10-INDEX	FIRST/CO-FIRST AUTHOR

Citation metrics from Google Scholar. Field-Weighted Citation Impact (FWCI) from Scopus/SciVal; FWCI above 1.0 indicates citation performance above the world average.

PROFESSIONAL APPOINTMENTS

Feb 2023 - present	**Senior Data Scientist (Cancer Data Science)**, Children's Medical Research Institute, Westmead
Jun 2026 - present	**Adjunct Lecturer**, Faculty of Medicine and Health, University of Sydney
Jun 2023 - Jun 2026	**Conjoint Associate Lecturer**, Faculty of Medicine and Health, University of Sydney
Mar 2020 - Feb 2023	**PhD Candidate (Cancer Data Science)**, Children's Medical Research Institute / University of Sydney
Jan 2019 - Mar 2020	**Data Scientist / Bioinformatician**, Children's Medical Research Institute, Westmead
Nov 2014 - Jan 2018	**Analyst Programmer**, Goldman Sachs, Melbourne Electronic Trading (GSET) division. Delivered the Shenzhen-Hong Kong Stock Connect project (Federation Award 2016) and established the India GSET clearing clients business flow.

QUALIFICATIONS

Mar 2020 - Feb 2023 — **Doctor of Philosophy (Cancer Data Science)**, University of Sydney / Children's Medical Research Institute

Thesis: Large-Scale and Pan-Cancer Proteogenomic Analyses with Machine Learning

- Sydney Cancer Partners PhD Scholarship recipient

Jan 2018 - Dec 2018 **Master of Business Analytics (First Class Honours)**, Melbourne Business School, University of Melbourne

- KPMG-MBS Data Challenge: 1st Prize Winner (Natural Language Processing)
- MBS Scholarship in Business Analytics; Co-President of Business Analytics Club

Jan 2011 - Jul 2014 **Bachelor of Computer Science (First Class Honours)**, Monash University

- Dux of Bachelor of Computer Science (highest overall ranking in graduating cohort)
- Bellamy Awards (top student in each academic year, 2011 and 2012)
- International Merit Scholarship (2011 - 2013)

Professional Training

2024 **Higher Degree Research Student Supervisor Training Course**, University of Sydney

GRANTS AND FELLOWSHIPS

Fellowships

2026 - 2028 **Cancer Institute NSW Early Career Fellowship**

$597,732 · Sole Chief Investigator · Awarded 2025

Improving Cancer Diagnosis and Prognosis via Generative Proteomic Foundation Model and Federated Learning

This competitive fellowship supports a three-year project developing proteomic foundation models integrated with federated learning for multi-hospital cancer research collaboration without sharing sensitive patient data.

Research Grants

2026 **ResetData AI-F1 Competition**, national GPU compute award

$1,000,000 (in-kind GPU compute) · Application lead

ProCan Multimodal Cancer Foundation Model

Led preparation of CMRI's competition entry, submitted under the Executive Director as institutional lead. The proposal seeks 64 NVIDIA H200 GPUs to train a multimodal foundation model integrating raw mass spectrometry files, proteomic data, histopathology images, and clinical metadata across more than 35,000 matched cancer tissue samples.

2024 - 2026 **Medical Research Future Fund (MRFF) Grant**

$993,500 · Associate Investigator

Targeting the Dysregulated Epigenome to Enhance Immunotherapy Response

Multi-institutional collaboration between Hudson Institute of Medical Research, CMRI, and Children's Cancer Institute. Contribution: computational biology expertise in multi-omic analysis and artificial intelligence.

Total competitive research funding contributed to: over $2.59 million (including $1 million in-kind GPU compute)

AWARDS AND HONOURS

2026	Cancer Research Network EMCR Outstanding Publication Award (Cai et al., Cancer Discovery, 2025)
2026	Cancer Bioinformatics Australia Symposium Travel Scholarship
2022	Sydney Cancer Partners PhD Scholarship
2020	CMRI Peter Rowe PhD Scholarship
2019, 2020	University of Sydney Innovation Challenge: 1st Prize ($7,500 each year)
2019	3rd Prize, ODIR-2019 International Computer Vision Competition (approx. $20,000)
2013	Dux of Bachelor of Computer Science, Monash University
2011 - 2013	International Merit Scholarship, Monash University
2011, 2012	Bellamy Award (Top Student), Monash University

PUBLICATIONS

Peer-Reviewed Journal Articles

Author name in bold. * denotes equal contribution. IF = journal impact factor. FWCI = Field-Weighted Citation Impact (Scopus/SciVal); FWCI above 1.0 indicates above world average.

1. **Cai, Z.**, Boys, E. L., Noor, Z., Aref, A. T., Xavier, D., Lucas, N., Williams, S. G., Koh, J. M., Poulos, R. C., Wu, Y., Dausmann, M., MacKenzie, K. L., Aguilar-Mahecha, A., Armengol, C., Barranco, M. M., Basik, M., Bowman, E. D., Clifton-Bligh, R., Connolly, E. A., … Reddel, R. R. (2025). Federated deep learning enables cancer subtyping by proteomics. *Cancer Discovery*, OF1-OF16.

   First author · IF 29.5 · FWCI 3.34 · 15 citations

   First application of federated deep learning to cancer proteomics, enabling model training across 7,500 cancer proteomes from multiple centres without sharing raw data.

2. Wu, Y., **Cai, Z.**, Cross, D., Noble, J. R., Prest, K., Littleboy, J., Cohen, S. B., Edlundh, B., Koh, J. M. S., Xu, R., Noor, Z., Bastami, M., Valentini, S., Richardson, L., Barthorpe, S., Arymanesh, N., Robinson, P. J., Hains, P. G., Garnett, M. J., … MacKenzie, K. L. (2025). Large-scale drug sensitivity, gene dependency, and proteogenomic analyses of telomere maintenance mechanisms in cancer cells. *Nature Communications*, 16(1), 11337.

   Co-author · IF 18.1 · FWCI 0.22 · 4 citations

   Contributed computational analyses for proteogenomic characterisation of telomere maintenance mechanisms.

3. Baião, A. R., **Cai, Z.**, Poulos, R. C., Robinson, P. J., Reddel, R. R., Zhong, Q., … Gonçalves, E. (2025). A technical review of multi-omics data integration methods: from classical statistical to deep generative approaches. *Briefings in Bioinformatics*, 26(4), bbaf355.

   Co-author · FWCI 11.60 · 197 citations

   International collaboration with University of Lisbon.

4. Osteil, P., Withey, S., Santucci, N., Aryamanesh, N., Pang, I., Salehin, N., … **Cai, Z.**, Wolvetang, E., & Tam, P. P. (2025). MIXL1 activation in endoderm differentiation of human induced pluripotent stem cells. *Stem Cell Reports*.

   Co-author · FWCI 0.76 · 4 citations

5. **Cai, Z.**, Apolinário, S., Baião, A. R., Pacini, C., Sousa, M. D., Vinga, S., … Gonçalves, E. (2024). Synthetic augmentation of cancer cell line multi-omic datasets using unsupervised deep learning. *Nature Communications*, 15(1), 1-12.

   First author · IF 18.1 · FWCI 3.38 · 47 citations

   Introduces MOSA (Multi-Omics Synthetic Augmentation), a generative AI approach using multi-view variational autoencoders. International collaboration with University of Lisbon.

6. **Cai, Z.**, Poulos, R. C., Aref, A., Robinson, P. J., Reddel, R. R., & Zhong, Q. (2024). DeePathNet: a transformer-based deep learning model integrating multi-omic data with cancer pathways. *Cancer Research Communications*, 4(12), 3151-3164.

   First author · IF 4.0 · FWCI 4.08 · 43 citations

   Novel transformer-based architecture integrating multi-omic data with cancer pathway knowledge for improved interpretability and prediction accuracy.

7. Gonçalves, E.*, Poulos, R. C.*, **Cai, Z.***, … Robinson, P., Zhong, Q., Garnett, M., & Reddel, R. R. (2022). Pan-cancer proteomic map of 949 human cell lines. *Cancer Cell*, 40(8), 835-849.

   Co-first author · IF 56.1 · FWCI 10.62 · 264 citations

   The world's largest pan-cancer proteomic resource at time of publication. Collaboration with Wellcome Sanger Institute (Cambridge, UK). Featured on ABC Radio and institutional media.

8. **Cai, Z.**, Poulos, R. C., Liu, J., & Zhong, Q. (2022). Machine learning for multi-omics data integration in cancer. *iScience*, 103798.

   First author · IF 4.5 · FWCI 6.04 · 344 citations

   Widely cited review for researchers entering the field of computational multi-omics integration.

9. Poulos, R. C., **Cai, Z.**, Robinson, P. J., Reddel, R. R., & Zhong, Q. (2022). Opportunities for pharmacoproteomics in biomarker discovery. *Proteomics*, 2200031.

   Co-author · FWCI 2.19 · 31 citations

   Invited Viewpoint article on pharmacoproteomics.

10. Guan, L., Tian, J., Cao, R., Li, M., **Cai, Z.**, & Shen, W. (2014). Barcode-like paper sensor for smartphone diagnostics: an application of blood typing. *Analytical Chemistry*, 86(22), 11362-11367.

    Co-author · IF 6.7 · FWCI 2.01 · 133 citations

    Interdisciplinary study integrating paper-based microfluidic diagnostics with smartphone technology for point-of-care blood typing.

TEACHING AND RESEARCH SUPERVISION

Higher Degree Research Supervision

Jul 2025 - present

Ms Fatemeh Mehdikhani, PhD candidate

Co-supervisor · Faculty of Medicine and Health, University of Sydney

Non-Invasive Diagnosis and Classification of Nevi and Early-Stage Melanoma

Provides guidance on computational methodology, machine learning model development, and data analysis approaches.

Thesis Examination

Oct 2025

Ms Rita Brito Gama, Master of Philosophy

External thesis examiner · University of Lisbon, Portugal

GAIN-DANN: A Domain-Adversarial Generative Model for Missing Data Imputation in Proteomics

Responsibilities included critical evaluation of the written thesis and participation in the oral defence. Ms Brito Gama successfully defended her thesis and was awarded her degree.

Mentorship of Researchers

CMRI

Dr Emma Boys, Medical Oncologist and PhD candidate

Guidance on machine learning methods for cancer classification research (Cancers of Unknown Primary). Co-authored publication in Cancer Discovery (2025).

CMRI

Dr Liz Connolly, Medical Oncologist and PhD candidate

Technical guidance on bioinformatic analysis, differential expression analysis, and survival analysis methods for prostate cancer research.

CMRI

Dr Di Xiao, Junior researcher, Cancer Data Science team

Support for integration into the team, guidance on computational platforms and analytical tools.

Formal Teaching Experience

2013 - 2014

Casual Tutor, Algorithms and Data Structures, Monash University

Delivered tutorials in computational subjects supporting student learning in algorithm design and implementation.

Staff Supervision (Forthcoming)

2026 - 2028

The Cancer Institute NSW Early Career Fellowship includes funding for a junior postdoctoral position. Expected to serve as primary supervisor, providing direction on computational approaches to cancer proteomics and generative AI methods.

CONFERENCE PRESENTATIONS AND INVITED TALKS

Oral Presentations

2026 — **Cancer Bioinformatics Australia Symposium**, Brisbane

Augmenting the Childhood Cancer Cell Map through Multi-Omic Data Synthesis and Predictive Modelling

2024 — **Advancing Multi-Omics into the Clinic Symposium**

2022 — **70th ASMS Conference on Mass Spectrometry and Allied Topics**

2021 — **KCA Precision Medicine for Childhood Cancer Symposium** **Invited**

2021 — **Combined ABACBS and Phylomania Hybrid Conference**

Poster Presentations

2023 — ABACBS Annual Conference

2023 — International Conference on Intelligent Systems for Molecular Biology (ISMB)

2022 — Human Proteome Organisation (HUPO) World Congress

2022 — ABACBS Annual Conference

Institutional Seminars and Media

2023 — **CMRI All Staff Seminar**
Presented research on computational approaches to cancer multi-omics data analysis to the broader institute community.

2022 — **ABC Radio interview**
Interview on the Cancer Cell pan-cancer proteomic map publication, communicating the significance of the research to a public audience.

SERVICE AND LEADERSHIP

Internal Service

2023 - present — **Leadership of Cancer Data Science Journal Club**, CMRI
Managed and hosted this monthly forum, bringing together 10 to 15 researchers from across the ProCan research initiative (Cancer Data Science, Software Engineering, and Oncology teams). Facilitates cross-team discussion, knowledge exchange, and critical evaluation of recent publications in cancer data science and machine learning.

2023 - present — **Participation in CMRI Research Seminars and Scientific Meetings**
Active engagement with presentations across diverse research areas, contributing to discussions that strengthen the collaborative research environment at CMRI.

Peer Review

2023 - present — Reviewed approximately five manuscripts per year since 2023, totalling over ten reviews. Journals include Nature Communications, Genome Biology, and Briefings in Bioinformatics, among others.

Grant Assessment

2025 — **Innovation and Technology Commission**, Hong Kong
Invited external expert reviewer for competitive national funding scheme.

2025 — **Dutch Research Council (NWO)**, Netherlands
Invited external expert reviewer for competitive national funding scheme.

INTERNATIONAL AND NATIONAL COLLABORATIONS

Cambridge, UK — **Wellcome Sanger Institute**
Ongoing collaboration in cancer proteomics and drug screening. Resulted in the landmark Cancer Cell publication (2022) and the Nature Communications telomere maintenance study (2025).

Lisbon, Portugal — **University of Lisbon**
Ongoing collaboration in multi-omics data integration and generative AI methods. Produced the Nature Communications MOSA publication (2024) and the Briefings in Bioinformatics technical review (2025).

Australia — **Sydney Cancer Partners and multi-institutional collaborations**
Contributing computational biology expertise to the MRFF-funded project Targeting the Dysregulated Epigenome to Enhance Immunotherapy Response (Hudson Institute of Medical Research, CMRI, and Children's Cancer Institute).

KEY TECHNICAL SKILLS

Machine learning and AI — Deep neural networks, transformer architectures, variational autoencoders, generative models, federated learning, and multi-view integration methods.

Bioinformatics and multi-omics — End-to-end proteomic data analysis (data QC, peptide-to-protein rollup, pre-processing, differential expression, pathway analysis, survival analysis), whole exome/genome sequencing (germline/somatic mutations, copy number variations, structural variants), and single-cell RNA-seq analysis.

Programming and software — Python, R, PyTorch, SQL, C++, Linux, Perl, and bioinformatics software suites.

Research communication — Scientific writing for high-impact journals, conference presentations, peer review, and public engagement.